Explaining Heart Disease Predictions:
A Comparison of SHAP and Partial Dependence Plots

Final Report

Name	Felix Mrak
Student no.	s1145815
Date	April 2026

Code: https://github.com/felixleopold/xai-heart-disease

1 Introduction

Machine learning is used more and more in healthcare, for example to predict whether a patient is at risk of heart disease. These models can be quite accurate, but they often work as **black boxes**, making it hard to understand why they make a certain prediction. This is a problem, especially because the EU AI Act classifies medical AI systems as high-risk and demands transparency and human oversight [3]. The GDPR also requires that people can get an explanation when an automated decision affects them [4].

Another concern is **bias**. A model might rely too much on demographic features like sex or age in ways that are not medically justified. Biases can enter at every stage of the ML pipeline and are especially dangerous in healthcare because they directly affect patient outcomes [6]. Explainable AI (XAI) methods can help here because they show us which features the model uses and whether that makes sense. But different XAI methods show different things. Local explanations tell us about one specific patient, while global explanations show general patterns across the whole dataset. That is why we use one of each and compare what they find on the same healthcare dataset.

2 Research Question

> *How do SHAP and Partial Dependence Plots compare in revealing model behaviour and potential demographic bias in a heart disease prediction model?*

The reason for comparing these two is that SHAP explains individual patients (local), while Partial Dependence Plots show the average effect of a feature across all patients (global). If both methods point to the same patterns, we can be more confident about how the model works and whether it treats different groups of patients fairly.

3 Methods

3.1 Dataset

We use the Heart Disease dataset that combines several sub-datasets (Cleveland, Hungarian, Switzerland, Long Beach V) from the UCI Machine Learning Repository [5]. This merged version has 1025 patients and 13 features, with a binary target that says whether the patient has heart disease or not (526 positive, 499 negative). The features include clinical measurements like blood pressure, cholesterol, and maximum heart rate, but also demographic ones like sex and age. Because it has these demographic features the dataset is well suited for checking whether the model might be biased. Sex is coded as 0 (female) and 1 (male).

3.2 Model

We trained a Random Forest with 200 trees using scikit-learn, with an 80/20 train-test split. The model got 99% accuracy on the test set. This seems very high and is probably inflated, but still works well for exploring the two XAI methods.

3.3 SHAP

SHAP (SHapley Additive exPlanations) is based on Shapley values from game theory [1]. The idea comes from a fair division problem: if a group of players together earn some payoff, how much should each player get? SHAP applies this to predictions, it splits the prediction among the input features based on how much each one contributed. So if a patient is predicted to have heart disease with probability 0.99 while the average is 0.52, SHAP tells us which features are responsible for that difference. It works by looking at every possible subset of features, checking how much the prediction changes when a feature is added to the coalition, and then averaging over all coalitions. SHAP is a local method, so it explains one patient at a time. A waterfall plot shows the breakdown for one patient, and a beeswarm plot shows all patients at once (Figure 1).

3.4 Partial Dependence Plots

A Partial Dependence Plot (PDP) answers a different question. On average, what happens to the prediction when we change one feature? [2] You take every patient, force one feature (for example age) to a specific value, keep everything else the same, then predict, and average over all patients. Repeat for many values of age and you get a curve that shows the average effect of that feature. One important limitation is that PDPs assume the feature we change is independent of the other features. When features are correlated this can produce misleading curves. We also made a 2-D PDP for age $\times$ sex to check for interaction effects.

3.5 Comparison and Evaluation

SHAP handles feature interactions through the Shapley framework, while PDPs assume features are independent. PDPs are simpler to explain to non-technical people, but the independence assumption can be misleading when features are correlated. To evaluate the methods we check whether the patterns they show are medically plausible and whether the two methods agree on the role of sex.

4 Code Description

The code is in a Jupyter notebook (`assignment.ipynb`) using pandas, scikit-learn, matplotlib, and shap. It loads the data, trains a Random Forest, computes SHAP values with TreeExplainer, and produces a waterfall, beeswarm, bar, and dependence plot. For PDPs it uses `PartialDependenceDisplay` to create 1-D PDPs for age, thalach, chol, and oldpeak, and a 2-D PDP for age $\times$ sex. All code is on GitHub[1].

5 Results and Discussion

5.1 SHAP Results

The SHAP bar plot shows that the most important features are chest pain type (cp, mean $|\phi| = 0.11$), number of major vessels (ca, 0.10), and thalassemia type (thal, 0.10). This makes sense because these are known risk factors for heart disease. Sex ranks near the bottom with mean $|\phi|$ of only 0.03, so on average it does not seem like a major driver. The beeswarm plot (Figure 1) shows this in more detail. For cp, ca, and thal, high feature values push toward disease, while for thalach (max heart rate) higher values push away from disease.

[1] https://github.com/felixleopold/xai-heart-disease

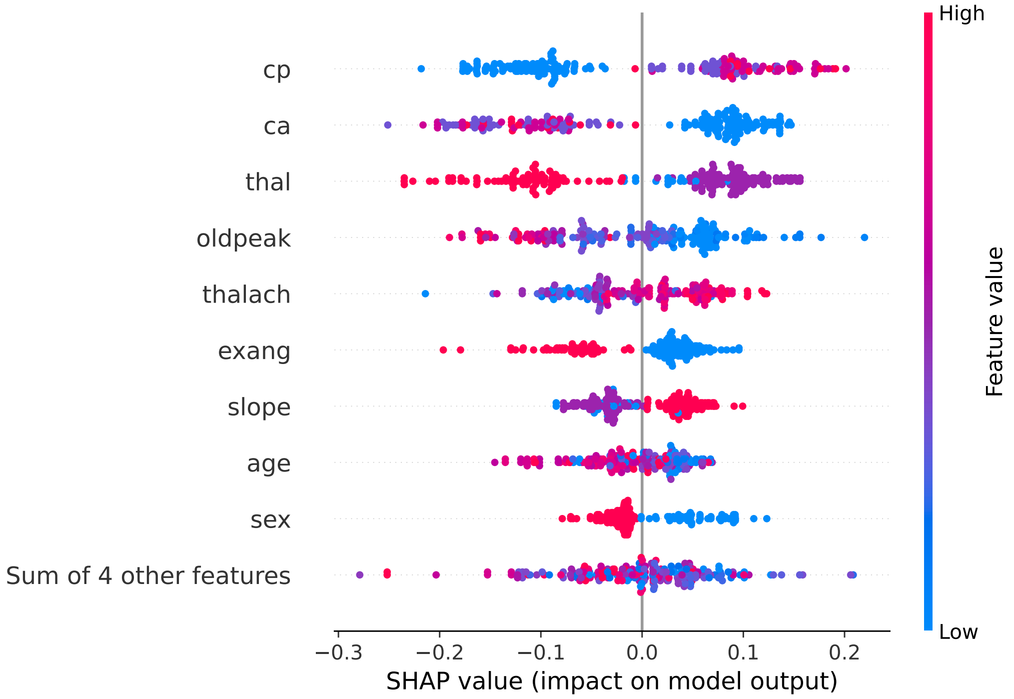

Figure 1: SHAP beeswarm plot showing feature contributions for all test patients.

But the dependence plot for sex (Figure 2) tells a different story. For female patients (sex = 0), SHAP values are almost all positive (+0.02 to +0.12), pushing predictions toward disease. For males (sex = 1), values are mostly negative (-0.02 to -0.08), pushing away from disease. So even though the average importance of sex is low (the positive and negative values cancel out), it consistently pushes in opposite directions for the two groups. This could reflect real clinical differences or a bias in the data.

Figure 2: SHAP dependence plot for sex, coloured by age.

5.2 PDP Results

The 1-D PDPs (Figure 3) show some expected and some surprising patterns. The thalach PDP rises from about 0.42 to 0.56 as heart rate increases. The oldpeak PDP drops from 0.60 to 0.35 as ST depression increases.

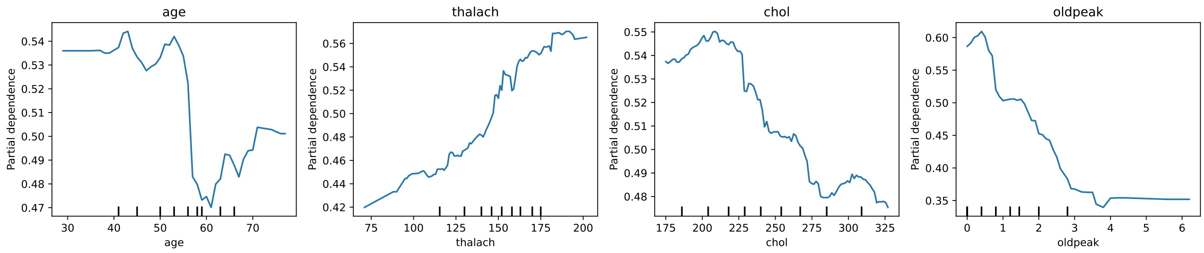

Figure 3: 1-D Partial Dependence Plots for four features.

The age PDP is surprising. Risk stays around 0.54 for younger patients and drops to about 0.47 after age 55. This is counterintuitive because heart disease risk generally increases with age. The likely explanation is that age correlates with other features like thalach and oldpeak, so when the PDP forces age to a high value but keeps the other features unchanged, it creates unrealistic patient profiles.

The 2-D PDP (Figure 4) shows an interaction effect. For females (sex close to 0) the partial dependence is higher (∼0.54–0.57) than for males (∼0.46–0.52). This is consistent with the SHAP results: both methods agree the model assigns higher risk to female patients.

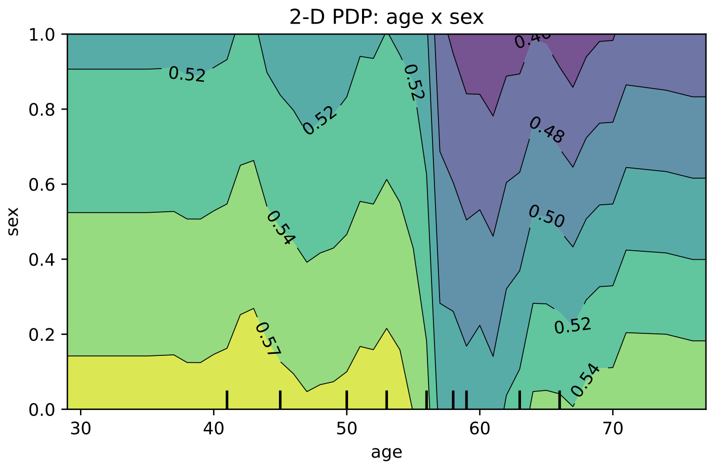

Figure 4: 2-D PDP for age × sex.

5.3 Pitfalls and Ethical Considerations

Both methods have pitfalls. PDPs assume features are independent, which is violated in our data, leading to the weird age pattern. A doctor looking at that PDP might wrongly conclude the model thinks older patients are safer. SHAP handles interactions better but is harder to explain to non-technical users.

The 99% accuracy is also suspicious. The dataset has known duplicates from merging four sub-datasets, so the model is probably overfitting. This does not invalidate our XAI analysis, but the model should not be used as a clinical predictor.

Ethically, the fact that sex influences predictions needs careful thought. Sex differences in heart disease are real, so some influence could be medically justified. But if the model learned from biased data where women were historically underdiagnosed, it could reinforce those disparities. Before deploying something like this, the predictions should be checked against actual clinical outcomes for different demographic groups.

References

[1] Lundberg, S. M. & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30.

[2] Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5), 1189–1232.

[3] Panigutti, C., Hamon, R., Hupont, I., Fernandez Llorca, D., Fano Yela, D., Junklewitz, H., ... & Gomez, E. (2023). The role of explainable AI in the context of the AI Act. In *2023 ACM Conference on Fairness, Accountability, and Transparency (FAccT '23)* (pp. 1139–1150). ACM.

[4] European Parliament and Council of the European Union (2016). Regulation (EU) 2016/679 (General Data Protection Regulation). *Official Journal of the European Union*, L 119, 1–88.

[5] Janosi, A., Steinbrunn, W., Pfisterer, M. & Detrano, R. (1989). Heart Disease [Dataset]. UCI Machine Learning Repository. Kaggle mirror: https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset

[6] Gichoya, J. W., Thomas, K., Celi, L. A., Safdar, N., Banerjee, I., Banja, J. D., ... & Purkayastha, S. (2023). AI pitfalls and what not to do: mitigating bias in AI. *The British Journal of Radiology*, 96(1150), 20230023.